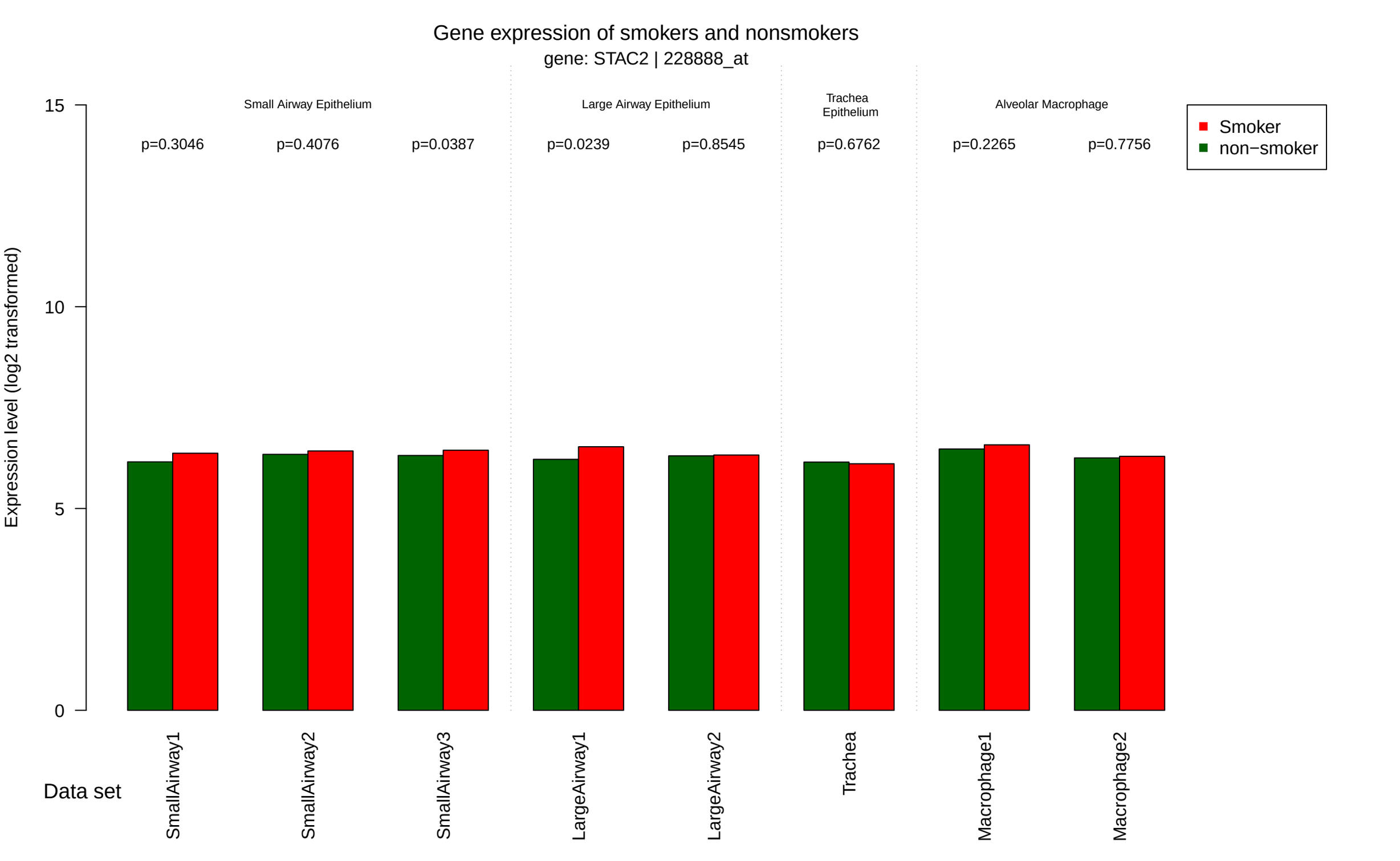
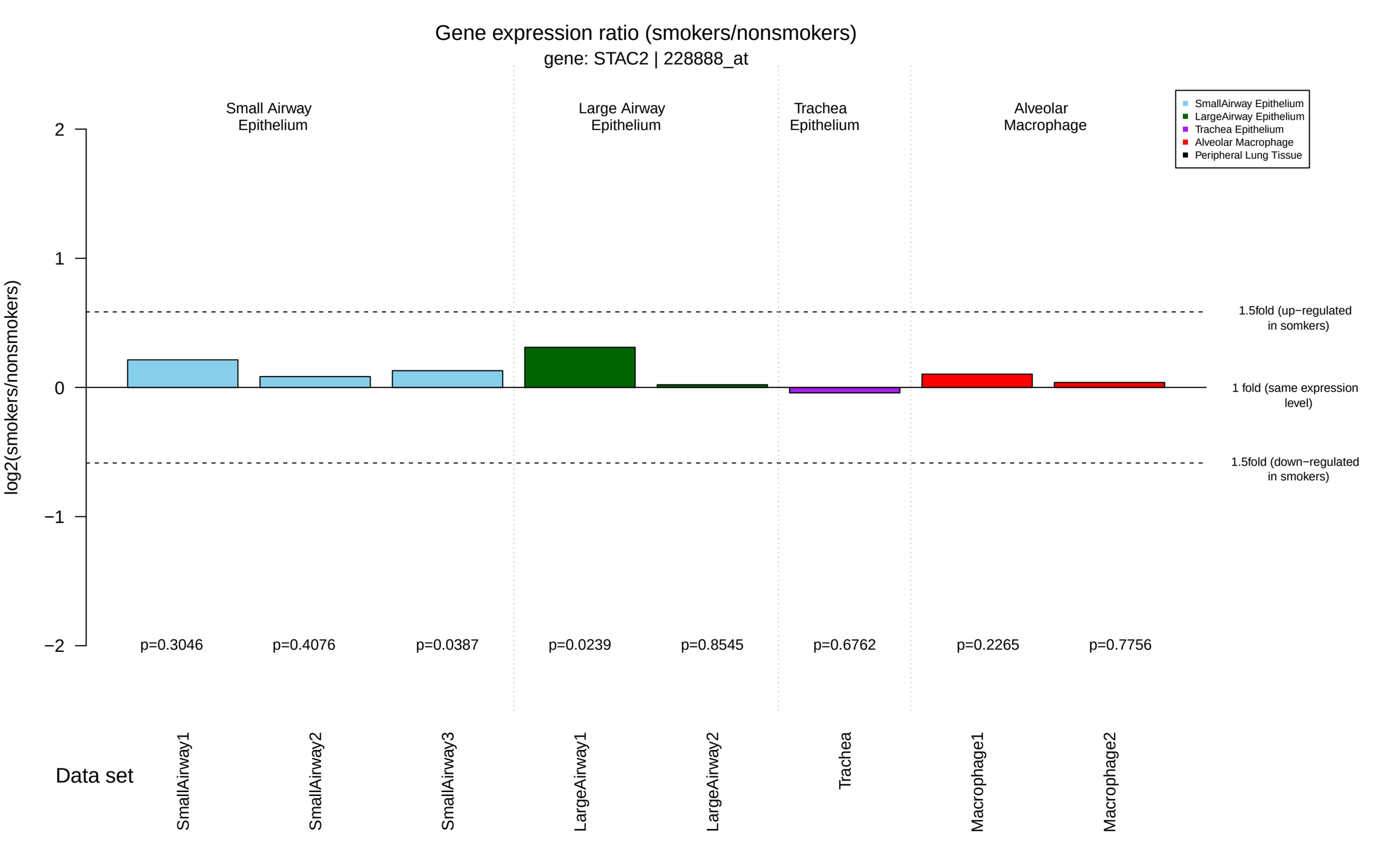
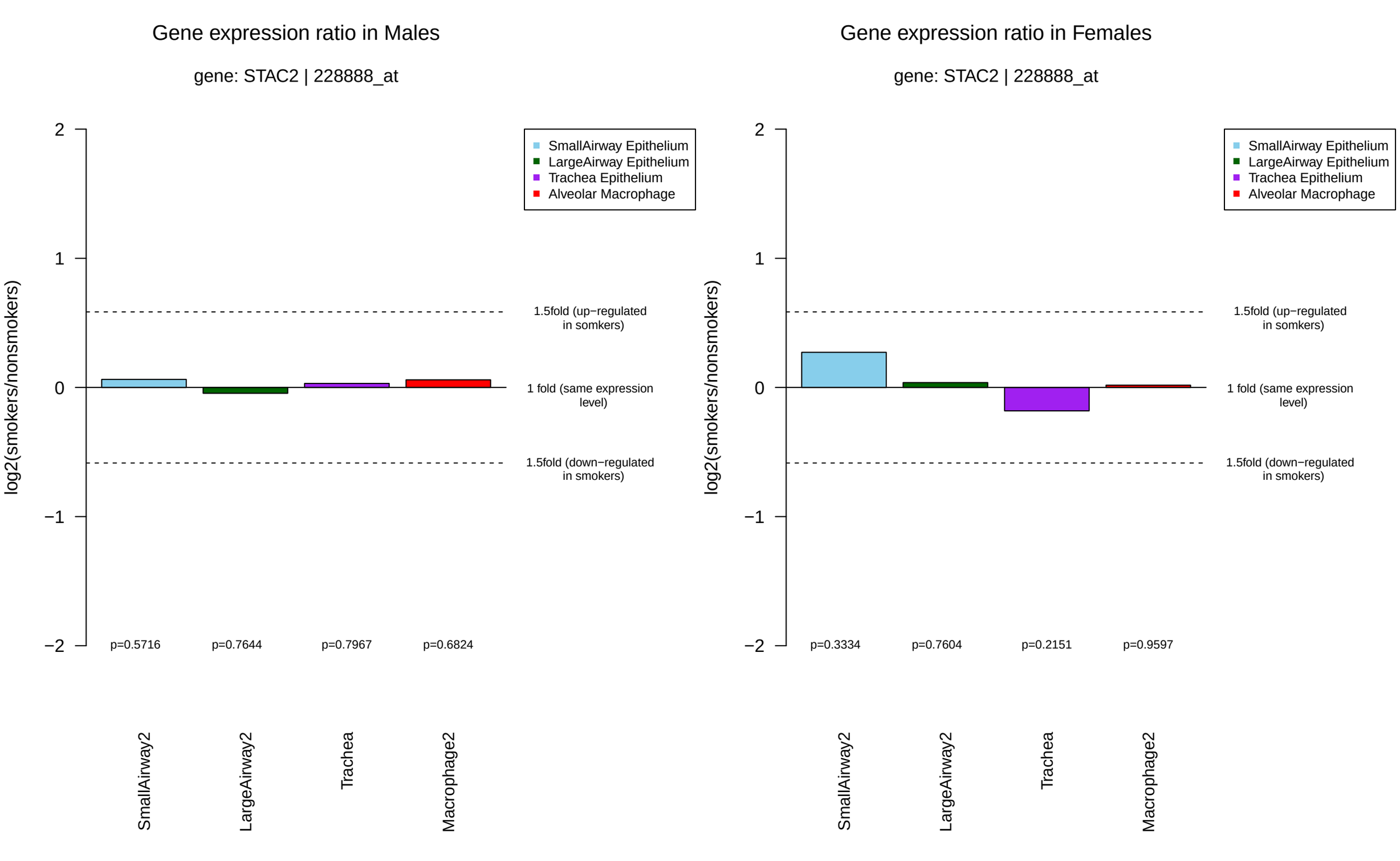
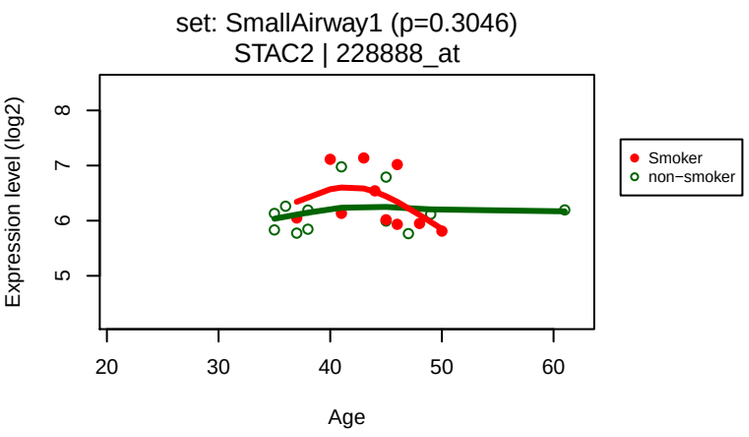
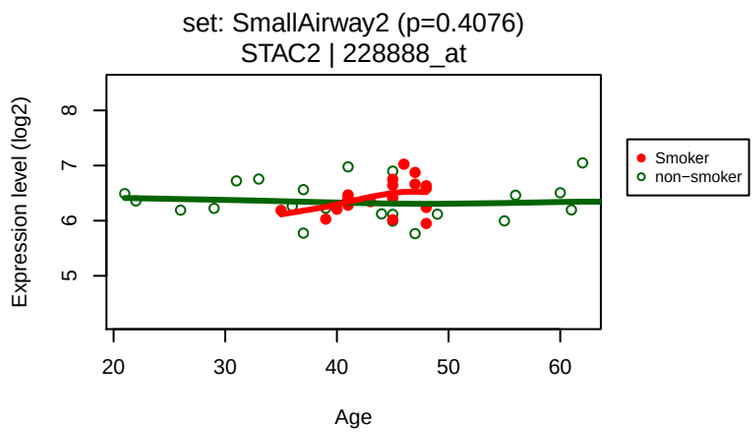
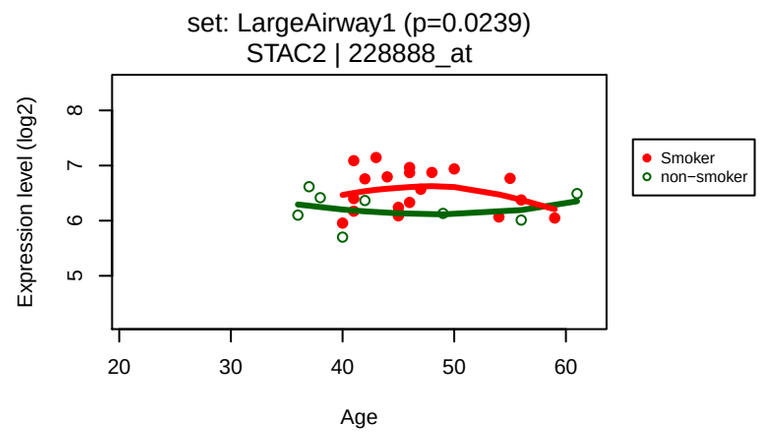

Gene expression of smokers and nonsmokers
gene: STAC2 | 228888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3046
p=0.4076
p=0.0387
p=0.0239
p=0.8545
p=0.6762
p=0.2265
p=0.7756
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

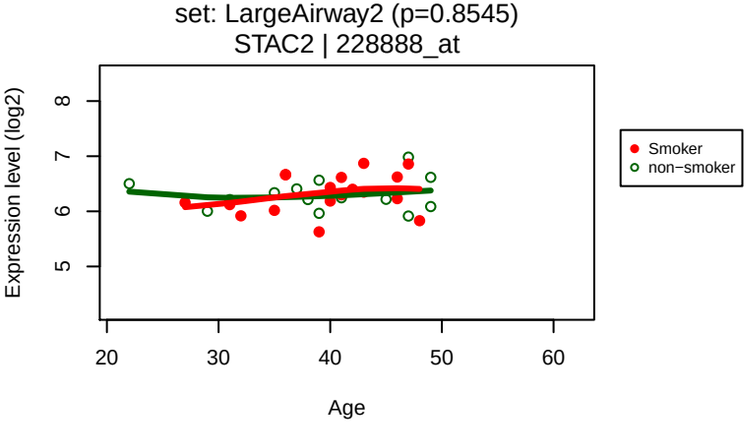

Gene expression ratio (smokers/nonsmokers)
gene: STAC2 | 228888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3046
p=0.4076
p=0.0387
p=0.0239
p=0.8545
p=0.6762
p=0.2265
p=0.7756
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

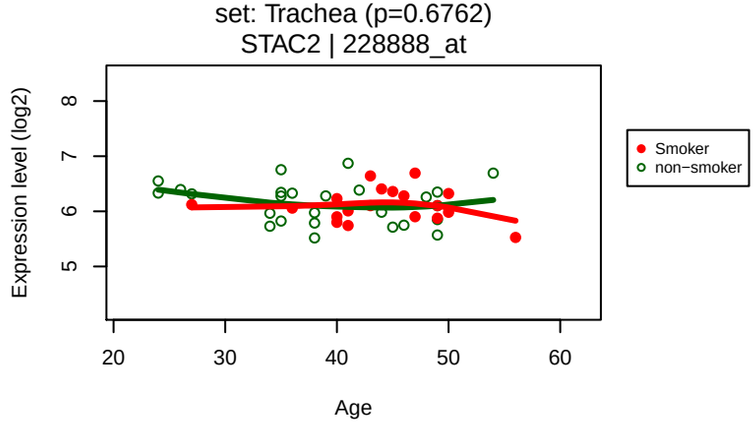

Gene expression ratio in Males
gene: STAC2 | 228888_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5716
p=0.7644
p=0.7967
p=0.6824
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAC2 | 228888_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3334
p=0.7604
p=0.2151
p=0.9597
SmallAirway2
LargeAirway2
Trachea
Macrophage2

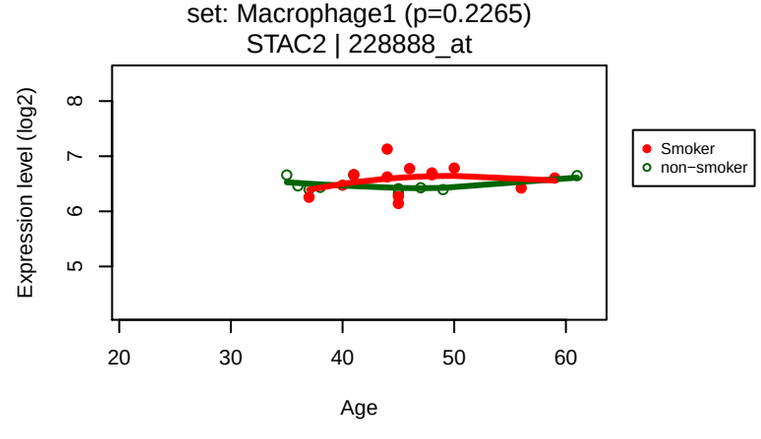

set: SmallAirway1 (p=0.3046)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker

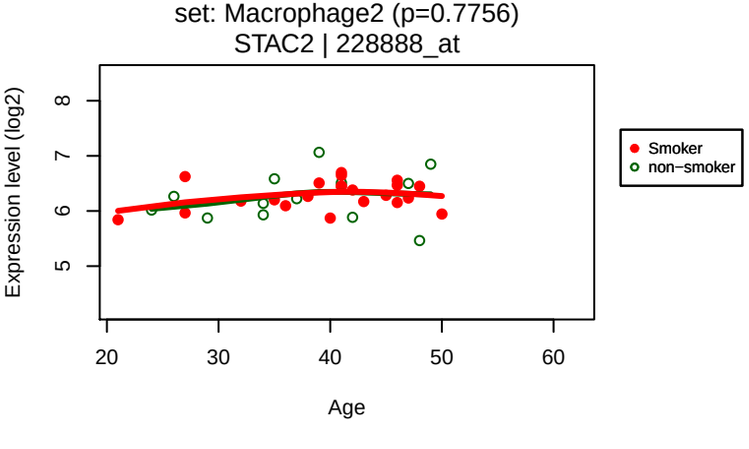

set: SmallAirway2 (p=0.4076)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker

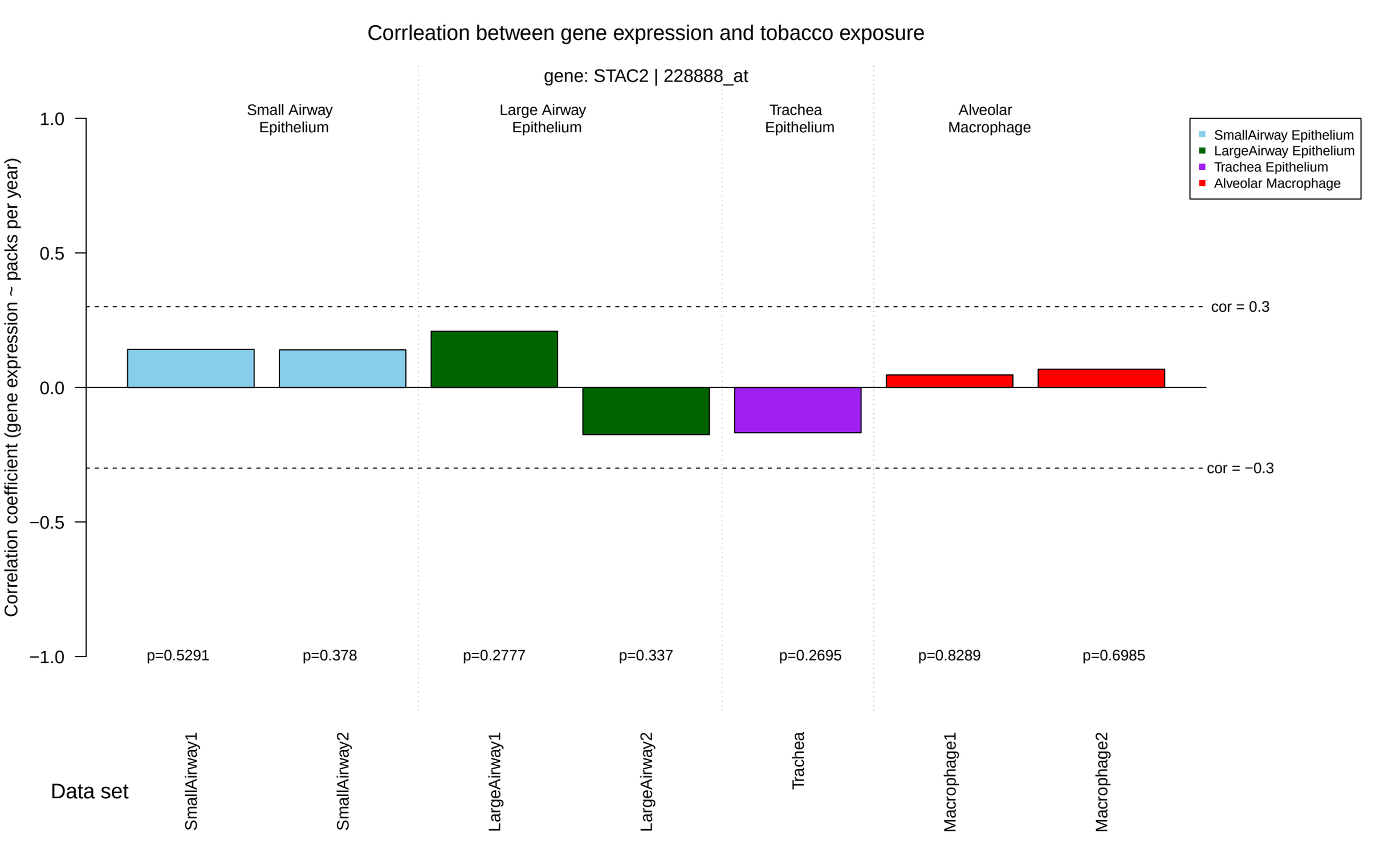

set: LargeAirway1 (p=0.0239)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8545)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6762)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2265)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7756)
STAC2 | 228888_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAC2 | 228888_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5291
p=0.378
p=0.2777
p=0.337
p=0.2695
p=0.8289
p=0.6985
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2